

§2.3

Multiplication and Division Problem Types

Four structures, and the unknown that names each subtype

Today's Goal

After completing this section, you should be able to:

- Differentiate between the different types of multiplication and division problems based on context
- Understand how the different types of multiplication and division problems influence the models and strategies used

LT.5 Multiplication and Division Problem Types

I can identify by name and create different types of story problems (e.g., Equal Groups (Multiplication)) related to multiplication and division.

Four Types, Three Subtypes Each

Table 2.3.1: Multiplication and Division Problem Types

Type	Subtypes		
Equal Groups	Multiplication	Partitive Division	Quotative Division
Array	Multiplication	Partitive Division	Quotative Division
Area	Multiplication	Partitive Division	Quotative Division
Compare	Multiplication	Partitive Division	Quotative Division

Every type has three subtypes, and the subtype is named by **which quantity is unknown**.

Equal Groups Problems

Definition 2.3.2

An **Equal Groups problem** involves a set of objects divided into groups in which each group has the same number of objects.

Components: **number of groups**, **group size**, **product** (the total across all groups).

number of groups	×	group size	=	product
product	÷	number of groups	=	group size
product	÷	group size	=	number of groups

The Three Equal Groups Subtypes

Table 2.3.3: Equal Groups Problem Subtypes

Equal Groups (Multiplication)	Equal Groups (Partitive Division)	Equal Groups (Quotative Division)
There are three bags with five apples in each bag. How many apples are there in all?	If 15 apples are shared equally into three bags, how many apples will be in each bag?	If 15 apples are to be packed five apples to a bag, how many bags are needed?
$3 \times 5 = ?$	$3 \times ? = 15$	$? \times 5 = 15$
	$15 \div 3 = ?$	$15 \div 5 = ?$

The student videos we will watch are from *Children's Mathematics* (Cognitively Guided Instruction, Thomas Carpenter and colleagues).

Grandma's Cookies

Grandma has 3 plates with 6 cookies on every plate. How many cookies does Grandma have?

- a. **Face Partner A:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.
Face Partner B: How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b. ▶ **Play: a student solves this problem** smarturl.it/cm4.1
- c. ▶ **Play: another student solves a similar problem** smarturl.it/cm4.5
- d. What type (and subtype) of multiplication or division problem is this an example of?

Mr. Gomez's Cupcakes

Mr. Gomez had 20 cupcakes. He put the cupcakes into 4 boxes so that there were the same number of cupcakes in each box. How many cupcakes did Mr. Gomez put in each box?

- a. **Face Partner B:** Use the cubes at your table to show your face partner how a child may use direct modeling to solve this problem.
Face Partner A: How else might a child go about solving this problem in a way that demonstrates making sense of the problem?
- b.  **Play: a student solves this problem** smarturl.it/cm4.4
- c. What type (and subtype) of multiplication or division problem is this an example of?

Coach Brown's Baseballs

Coach Brown has 72 baseballs. If 8 baseballs fit in a box, how many boxes can Coach Brown fill?

a. **Face Partner A:** Explain to your face partner why you don't want to use cubes to solve this problem with direct modeling.

Face Partner B: How else might a child go about solving this problem in a way that demonstrates making sense of the problem?

b.  **Play: a student solves this problem** smarturl.it/cm4.2

c. What type (and subtype) of multiplication or division problem is this an example of?

Which Unknown Names the Subtype?

Problem	Given	Unknown	Subtype
Grandma's cookies	3 groups, 6 in each	product	Equal Groups (Multiplication)
Mr. Gomez's cupcakes	20 in all, 4 groups	group size	Equal Groups (Partitive Division)
Coach Brown's baseballs	72 in all, 8 in each	number of groups	Equal Groups (Quotative Division)

Quotative division is also called **measurement division**: we use the given size of each group to *measure* the total number of objects and see how many groups there are.

Array Problems

Definition 2.3.6

An **Array problem** involves a set of objects arranged in a rectangular array.

Components: **number of rows**, **number of columns**, **product** (the total number of objects).

number of rows	×	number of columns	=	product
product	÷	number of rows	=	number of columns
product	÷	number of columns	=	number of rows

The Three Array Subtypes

Table 2.3.8: Array Problem Subtypes

Array (Multiplication)	Array (Partitive Division)	Array (Quotative Division)
There are three rows of apples with five apples in each row. How many apples are there?	If 15 apples are arranged into three equal rows, how many apples will be in each row?	If 15 apples are arranged into equal rows of five apples, how many rows will there be?
$3 \times 5 = ?$	$3 \times ? = 15$	$? \times 5 = 15$
	$15 \div 3 = ?$	$15 \div 5 = ?$

Why Does $5 \times 4 = 4 \times 5$?

A child reads Grandma's cookies as **three groups of six**, never six groups of three. The equal groups model is not **symmetric**. Arrays show children why the two products agree.



5 rows of 4 chairs



4 rows of 5 chairs

Same chairs, rearranged: the total does not change, so $5 \times 4 = 4 \times 5$.

Area Problems

Definition 2.3.12

An **Area problem** involves representing the area of a rectangle using multiplication or division.

Rectangles only. Components: **length**, **width**, **product** (the area of the rectangle).

length	×	width	=	area
area	÷	length	=	width
area	÷	width	=	length

The Three Area Subtypes

Table 2.3.13: Area Problem Subtypes

Area (Multiplication)	Area (Partitive Division)	Area (Quotative Division)
What is the area of a three inch by five inch rectangle?	A rectangle has an area of 15 square inches. If the length is three inches long, how long is the width?	A rectangle has an area of 15 square inches. If the width is five inches long, how long is the length?
$3 \times 5 = ?$	$3 \times ? = 15$	$? \times 5 = 15$
	$15 \div 3 = ?$	$15 \div 5 = ?$

Which Side Is the Length?

**A 3 inch by 5 inch rectangle:
which side is the length?**

Either one. Length and width are interchangeable, so for Area problems the two division subtypes are too.

Table 2.3.13 tells them apart by the given side. **Length given:** Partitive Division.
Width given: Quotative Division.

Multiplicative Compare Problems

Definition 2.3.16

A **Multiplicative Compare problem** involves a comparison between two quantities in which one quantity is described as a multiple of the other quantity.

Components: **smaller** quantity, **bigger** quantity, **scale factor** (how many times larger).

scale factor	×	smaller quantity	=	bigger quantity
bigger quantity	÷	smaller quantity	=	scale factor
bigger quantity	÷	scale factor	=	smaller quantity

Multiplicative Compare Subtypes

Table 2.3.17: Multiplicative Compare Problem Subtypes

Multiplicative Compare (Multiplication)	Multiplicative Compare (Partitive Division)	Multiplicative Compare (Quotative Division)
A blue hat costs \$5. A red hat costs 3 times as much as the blue hat. How much does the red hat cost?	A red hat costs \$15 and that is 3 times as much as a blue hat costs. How much does a blue hat cost?	A red hat costs \$15 and a blue hat costs \$5. How many times as much does the red hat cost as the blue hat?
$3 \times 5 = ?$	$3 \times ? = 15$	$? \times 5 = 15$
	$15 \div 3 = ?$	$15 \div 5 = ?$

Write Three Problems From One Fact

Work with your group to use the math fact $4 \times 3 = 12$ (or a related math fact) to create

- a. an “Array (Partitive Division)” problem.
- b. an “Area (Quotative Division)” problem.
- c. a “Multiplicative Compare (Multiplication)” problem.

Check each problem against its number facts: which quantity did you leave unknown?